Proposal Feedback - English Translation

Translation of the instructor feedback on the proposal

Grade: 92/100 - Approved

Grade Breakdown

Component	Score
Research Question	**40/40**
Database Description	**27/30**
Methodological Robustness	**25/30**

What Was Done Well

Both data sources are official government sources - Ministry of Education data obtained through the government data portal, and the Central Bureau of Statistics (CBS) socioeconomic index for local authorities - and merging them constitutes genuine data work rather than simply analyzing a single file.

The research question is specific, measurable, and answerable using the sources you identified. The target audience you defined - education policymakers, heads of local authorities, and the Ministry of Education - is realistic, with a clear path from the findings to decisions regarding budget allocation and targeted interventions.

The strongest indication of the quality of the proposal is that it is clear you actually opened and examined the files. You identified in advance that the two systems use different numbering methods for local authorities and provided a concrete example: Ofakim is coded as 315 in the education data and 31 in the CBS data. You also identified tested_students_n as a variable that should be examined because of noise caused by small sample sizes.

The row counts and variables were specified precisely, the columns are referred to by their actual field names, and the target variable is clearly defined.

The work plan is comprehensive: exploratory data analysis, correlations, train/validation/test splitting, clustering, a regression track and a classification track with two competing models in each, and feature importance analysis designed to answer a well-formulated question - whether the socioeconomic index outweighs grade level and subject.

The clustering idea, in which you construct academic groupings based on the data itself and compare them with the official CBS clusters, is the most interesting part of the proposal.

What Needs to Be Corrected in the Final Report

1. Split the data by local authority, not by row.

The 3,070 rows are combinations of local authority x grade level x subject x year and they are derived from only approximately 265 local authorities. Therefore, the rows are not independent.

A random split at the row level would place the same local authority in both the training and test sets. As a result, the model would partially learn to memorize authority-level averages, and the reported accuracy would be inflated.

Use a grouped split, such as GroupShuffleSplit or GroupKFold, with the local authority as the grouping variable.

The results may become worse after doing this, and that is exactly the point: an honest evaluation is more valuable than an impressive-looking number.

2. Remove causal wording.

The question is currently phrased as: "How does the socioeconomic index affect academic achievement?"

The word "affect" implies causality, while the data are observational and cross-sectional.

Change the wording to something such as "is associated with" or "predicts", and add an explicit qualification regarding causality in the discussion section.

3. Address the ecological fallacy.

The data are aggregated at the local-authority level, while the research question refers to "the academic achievement of its students."

An association between a local authority's socioeconomic index and its average score does not justify any conclusion about an individual student.

Mention this limitation explicitly by name - ecological fallacy - and define the implications of this limitation. A single paragraph addressing this issue would substantially improve how defensible the conclusions are.

4. Make the deviation/outlier analysis a central theme.

The relationship between socioeconomic cluster and academic achievement in Israel has already been documented and published, including by the Ministry of Education itself.

RAMA (the National Authority for Measurement and Evaluation in Education) publishes multi-year achievement reports at the local-authority level, broken down by cluster, and the Knesset Research and Information Center has also published analyses of the socioeconomic index.

Therefore, showing that local authorities in higher socioeconomic clusters tend to achieve higher scores would largely reproduce a known finding.

What your project adds is the comparison between the clusters you generate from the data and the official clusters: Which local authorities perform better than their socioeconomic cluster would predict, and what characterizes them?

Lead with this question, and present the regression and classification models as mechanisms for producing the answer. Add a short paragraph citing the existing publications and explaining what your analysis contributes beyond them.

5. Define the classification threshold precisely.

"High versus low based on the median" is ambiguous.

Using one overall median would mainly encode which subject and grade level a row belongs to. As a result, the classifier would primarily learn the subject rather than something meaningful about socioeconomic status.

Instead, either calculate the median within each subject x grade-level group, or standardize the scores within each subject-grade combination before converting them into a binary target. State the rule explicitly.

6. Expand the predictor set.

After merging the datasets, you effectively have only three predictors: socioeconomic index, grade level, and subject.

A Random Forest does not have much information to work with in such a setup, and the resulting feature-importance comparison is almost predetermined.

The CBS local-authority files contain much more information beyond socioeconomic cluster, including population size, district, peripherality index, dependency ratio, and additional variables.

Adding several of these variables would make both the modeling and the feature-importance analysis more meaningful.

Also note that CLUSTER and VALUE INDEX are two different encodings of the same underlying measurement. Choose one of them for each use case and clearly specify which one you use.

7. Add cross-validation, a baseline, and weighting by the number of tested students.

Explain how hyperparameters will be tuned - using grouped cross-validation on the training set.

Report baseline models: for regression, predicting the mean; for classification, predicting the majority class.

In addition, because tested_students_n varies substantially between local authorities, consider either weighting observations according to this variable or excluding particularly small cells.

You already identified this issue during the exploratory analysis stage; continue addressing it during the modeling stage as well.

8. Specify the years and align them.

The education data span multiple years, while the CBS socioeconomic index is from the 2021 edition.

State explicitly which years are included and acknowledge the mismatch between the datasets, or restrict the analysis to years close to the year of the socioeconomic index.

9. Add an AI-use statement.

This was not required at the proposal stage, but Section 5 requires it in the final report.

Specify which AI tools were used, for what purposes, and which prompts were used.

Where Points Were Deducted

Research Question

No deduction - full score.

Database Description

3 points deducted:

- 1 point - CLUSTER and VALUE INDEX are two encodings of the same measurement, creating redundancy that was not addressed (see Section 6).
- 2 points - The effective predictor set is very limited, consisting of only three variables, and this limitation was not acknowledged in the proposal (see Section 6).

Methodological Robustness

5 points deducted:

- 4 points - The train/test split is performed at the row level rather than at the local-authority level, creating data leakage and an overly optimistic performance estimate (see Section 1).
- 1 point - Cross-validation, hyperparameter tuning, and a baseline were not specified (see Section 7).

No points were deducted for:

- the causal wording (Section 2),
- the ecological fallacy (Section 3),
- the lack of differentiation from the existing literature (Section 4),
- the definition of the binarization threshold (Section 5),
- failure to specify the years (Section 8), or
- the absence of an AI-use statement (Section 9).

However, all of these issues must be addressed in the final report.

Final note: The proposal grade is final and will not change as a result of subsequent improvements. However, failure to implement the recommendations above in the final report, without a good justification, will negatively affect the grade of the final report.